干式螺杆真空泵
使用说明书

Dry screw vacuum pump Instruction manual

敬请在使用前详阅本说明书以及相关文字资料。

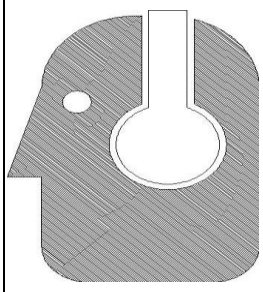
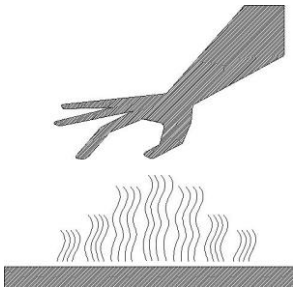
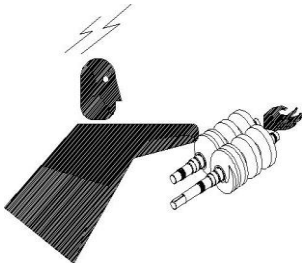
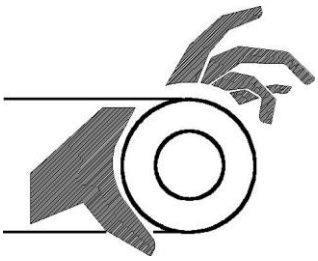
Please read the manual and relevant text before use.

江苏拓驰工程技术开发有限公司

Jiangsu Torchi Engineering & Technology Development Co., Ltd.

安 全 指 示
1. 在未阅读使用手册之前请不要操作设备。
2. 请使用足够的防护措施和防护设施，以防止安装和操作设备产生的不必要的损伤。

Please don't operate the equipment before reading the manual
Please use adequate protective measures and protective facilities to prevent unnecessary damage from installation and operating equipment.

警 告 Warning	注 意 Signify	警 告 Warning	警 告 Signify
			
须配置听觉保护装置 The auditory protection device must be configured	请勿将手置于设备表面 Do not place your hands on the surface of the device	将身体和衣物远离运转的设备 Keep the body and clothing away from the running equipment	请勿在无操作说明的情况下开动机器 Do not start the machine without operating instructions

注意 Warning

忽略这些警告标志会导致设备操作者或维护者严重的身体损伤。

Ignoring these warning signs can cause damage to the device's operator or maintainer.

安全事项 Safety matters

1. 连接在泵体上的管道不要有过大的重量。

Not on the pump body to have too much weight

2. 请不要在没有联轴罩的情况下开动设备，错误启动旋转方向会损伤设备。

Please do not start the device without the shaft cover, the error starts the rotating direction to damage the equipment

3. 进气端直接暴露于空气时，请不要运转设备。

Please do not run the equipment when the body is directly exposed to the air

4. 请注意，在给泵做任何维护时，务必先断开真空泵电源。

Please note that when you do any maintenance to the pump, you must disconnect the vacuum pump

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序言

本使用说明书叙述了LG系列、LGB系列螺杆真空泵的操作和维护说明和注意事项。强烈建议操作或维护本公司螺杆真空泵的人员仔细阅读本手册后再进行操作和维护，以延长产品的使用寿命。

Preface

This paper describes the operation and maintenance instructions and considerations of the lg series of LGB series. It is strongly recommended that people who operate or maintain the company's screw vacuum pump should read this manual carefully and then operate and maintain to extend the service life of the product.

1. 概述

1.1 概述

LG系列螺杆真空泵采用四级等导程的转子，LGB系列螺杆真空泵采用四级变导程的转子，这两个系列的螺杆真空泵的主、从动转子的采用相同的外廓型线，旋向相反，啮合间隙均匀，啮合线封闭，密封性好，抽气效率高。

螺杆真空泵是通过主、从动转子啮合转动使吸气段的封闭空间变大来吸入气体的，气体吸入泵腔后，随着泵的继续运转，在主、从动转子的螺旋啮合旋转的作用下，封闭空间从进气端向排气端移动，压力稳定上升，不存在中间排气现象；当封闭空间与排气口接通时，排气通道内的气体反冲至泵腔内，造成泵腔内的气体压力升高、温度升高，并带来气流噪声；随着泵的继续运转，封闭空间的体积开始逐渐变小，泵腔内的气体被不断地排出。本系列螺杆真空泵泵腔内无油润滑，可获得洁净真空。螺杆真空泵在抽吸有腐蚀性的气体时，需作防腐处理。

变导程螺杆真空泵特点：转子结构更复杂，精度要求更高，转子加工难度更高，加工工时更长；同时变导程螺杆真空泵运行功率较小、噪声较低、泵运行温度较低。

1.Overview of the situation

1.1 Overview

The rotor and LGB series screw vacuum pump is used in the rotor of the four-stage variable guide, the main and the same outer contour of the rotor vacuum pump of the two series of screw, the rotation of the rotor rotor, the rotation is opposite, the mesh is uniform, the mesh line is uniform, the mesh line is closed, the sealing is closed, the sealing is good, the gas efficiency is high.

The piston rotor engages the air inhalation of the gas, the gas inhaled the gas, the gas suction pump cavity, as the pump continues to operate, in the main, the spiral of the rotor of the rotor, the closed space moves from the wall to the exhaust end, the pressure is stable and rising, and there is no intermediate exhaust phenomenon; When the enclosed space is connected to the exhaust port, the gas in the exhaust channel is recoiled into the pump cavity, causing the gas pressure in the pump cavity to rise, the temperature rises, and the flow noise; As the pump continues to work, the volume of enclosed space starts to get smaller and smaller, and the gas in the pump is constantly expelled. This series screw vacuum pump cavity is not lubricated and can obtain a clean vacuum. The screw vacuum pump should be treated as an antiseptic treatment when it smokes a corrosive gas.

The characteristics of the rotary screw vacuum pump: the rotor structure is more complex, the accuracy requirement is higher, the rotor processing is more difficult and the processing time is longer. At the same time, the vacuum pump is smaller, the noise is low and the pump is low.

1.2 结构

●螺杆转子

螺杆转子材质为高等级球状石墨铸铁，由专用的数控设备精密制造，并通过精确的G1级动平衡试验。

●齿轮

齿轮是螺杆真空泵上最重要的部件之一，它能够使螺杆转子和转子之间保持固定的间隙。本系列螺杆真空泵采用高精度的斜齿轮结构，啮合均匀，稳定性好，运行噪声低。

●轴承

固定端轴承为双列角接触球轴承，延展端轴承为深沟球轴承。这些轴承均能够承受高速高负载运转，同时能够精确保持与齿轮和转子之间的间隙。

●轴封

密封结构采用迷宫式密封+充气密封+干式密封，这种密封可完全隔断润滑油腔与泵腔的通道，防止润滑油与被抽气体相互扩散、污染。

●润滑油

真空泵齿轮采用润滑油润滑，润滑油量应当添加至油镜中线稍上即可，但不可没过油镜。如果油量太少，油腔内进入流动的循环的润滑油油量就少，轴承和齿轮得不到有效的润滑，会因异常摩擦而受损；同时流动的润滑油少，不能有效地冷却齿轮与承轴，会影响真空泵的使用寿命，甚至出现故障。真空泵的油腔内配有专用的冷却系统冷却润滑油。所以，当对泵进行维护、保养时，请定期检查油量和油污染的状况，以决定是否需要增加或更换润滑油。

1.2 Structure

● Screw rotor

The screw rotor is made of high grade spherical graphite cast iron, which is made by special CNC equipment, and through the precise g1 stage dynamic balance test.

● Gear

The gear is one of the most important parts of the screw vacuum pump, which can keep the screw rotor and the rotor on a fixed gap. The series screw vacuum pump is used for high precision helical gear structure, the mesh is uniform, the stability is good, the running noise is low.

● Bearing

The fixed end bearing is the double column Angle contact ball bearing, the extension end bearing is the deep groove ball bearing. These bearings can withstand high speed and high load operations, while also able to keep the gap between the gears and the rotor.

● Shaft seal

The sealing structure adopts labyrinth seal + inflatable seal + dry seal, which can be completely separated from the channel of the oil cavity and pump cavity, preventing the oil and the gas from spreading and contamination.

● Oil

The vacuum pump gear is lubricated by lubricating oil, and the amount of lubricating oil should be added to the middle of the oil mirror, but it can't be no less oil. If the amount of oil is too small, the oil amount of the oil flowing in the oil cavity is low, and the bearings and gears are not effectively lubricated, and will be damaged by

abnormal friction. At the same time, the flow of lubricating oil is less, unable to effectively cooling the gear and the bearing shaft, which affects the life of the vacuum pump and even fails. The vacuum pump has a special cooling system for cooling oil. Therefore, when the pump is maintained and maintained, check the oil quantity and the condition of the oil stain regularly to determine whether to increase or replace the lubricating oil.

1.3 性能参数表 Performance parameter table

参数 \ 型号		LG50	LG70	LG100	LG150	LG200	LGB70	LGB110	LGB150	LGB220A	LGB300	LGB300A
抽 速 Pumping speed (50/60Hz)	m^3/Hr	180/216	250/300	360/430	540/650	720/850	250/300	360/430	540/650	792/950	1080/1290	1080/1290
	L/min	3000/3600	4200/5000	6000/7200	9000/10800	12000/14400	4200/5000	6000/7200	9000/10800	13200/15840	18000/21600	18000/21600
	L/S	50/60	70/84	100/120	150/180	200/240	70/84	100/120	150/180	220/264	300/360	300/360
极限压力 Extreme pressure (50/60Hz)	mbar	0.02 / 0.01										0.5 / 0.25
	Torr	0.015/0.0075										0.38 / 0.19
	Pa	2 / 1										50 / 25
功率(kW)50/60Hz		5.5/7.5	7.5/7.5	11/15	15/15	22/22	5.5/7.5	7.5/11	11/15	15/18.5	22/22	15/18.5
转速(RPM)50/60Hz		2900 / 3550										
进气口径 Inlet(mm)		50	65	80	90	100	50	65	80	90	100	90
出气口径 Outlet(mm)		40	40	65	65	65	40	40	65	65	65	65
齿轮腔油量 Gear cavity oil(L)		0.85	0.85	1.4	2	2	0.85	0.85	1.4	2	2	2
冷却水量 Cooling water quantity	L/min	2.5	2.6	2.8	3	4	2.5	2.5	2.6	2.8	3.5	2.8
	Gal/min	0.67	0.7	0.75	0.8	1.07	0.67	0.67	0.7	0.75	1.07	0.75
重量 weight（kg）		295	350	480	520	680	295	350	480	520	680	520

注意:(1)以上列出的油量仅供参考，通常为氟化物和矿物质油。设备出厂时配有泵油和润滑脂油。

(2)上述冷却水量是基于环境温度30℃，冷却水进水温度20℃的用量，使用循环水会有所不同，水压0.15~0.2MPa。

(3)LG螺杆真空泵是等导程的螺杆真空泵系列；LGB螺杆真空泵是变导程的螺杆真空泵系列。

(4)表上所示进、排气通径为单泵通径，非单泵设备的进、排气通径详见设备外形图。

Note：

(1) That the amount of oil listed above is for reference only, usually for fluoride and mineral oil. The equipment is equipped with pump oil and grease oil.

(2) The above cooling water is based on the environmental temperature of 30℃, the amount of water inlet temperature of 20℃, the use of the circulating water will be different, the water pressure is 0.15~ 0.2mpa..

(3) LG screw vacuum pump is the screw vacuum pump series of the same guide. LGB screw vacuum pump is the screw vacuum pump series of variable guide.

(4)The inlet diameter and the exhaust diameter is single pump diameter, and the non-single-pump device is in the input and exhaust diameter of the device.

1.4、清洗

● 冷却吹扫

冷却吹扫用于冷却转子和泵体。在泵运转过程中，需要冷却吹扫来冷却排出气体产生的压缩热量。**气体由进气口进入泵腔**，经过转子转动压缩并传送到排出端，气体被压缩产生的热量加热。该热量在排出端会达到100～200℃，所以我们需要使用冷却吹扫来降低这个温度。在大部分情况下，我们使用干燥、清洁的空气作为冷却吹扫物质，这是我们的标准吹扫物质。使用大气作为吹扫物质时，需要在进气端配置空气过滤器。冷却吹扫会占用泵的抽速，使真空泵的实际有效抽速下降，吹扫量越大，影响越明显。

1.4 Cleanout

● Cool-swept

The cooling blow is used to cool the rotor and pump. In the operation of the pump, the cooling blow is needed to cool the compressed heat generated by the gas. The gas is entered into the pump cavity by the pipe, and the rotor is compressed and transmitted to the discharge end, and the gas is heated by the heat generated by the compression. The heat will reach 100~ 200x at the end of the discharge, so we need to use the cooling blow to reduce the temperature. In most cases, we use dry, clean air as a cooling purge, which is our standard purge. When using the atmosphere as an purge, you need to configure the air filter at the back of the box. The cooling blow will take up the pumping speed of the pump, and the actual effective pumping of the vacuum pump will decrease, and the increase of the blow will be more obvious.

⑴ 吹扫量：

型　　号	用量(m^3/min)	说　　明
LG-50	0.55	用量随泵工作真空度的变化有所差异。
LG-70	0.6	
LG-100	0.75	
LG150	0.85	
LG200	1	
LGB70	0.45	
LGB110	0.55	
LGB150	0.7	
LGB220A	0.8	
LGB300	0.95	
LGB300A	0.8	

① 标准：通过空气过滤器的大气；

② 使用热交换器 ： 使用热交换器将冷却的排出气体循环利用；

③ 使用N_2或惰性气体作为吹扫介质。

(1) Blow sweep:

Type	Number(m^3/min)	Brightness
LG-50	0.55	The amount of consumption varies
LG-70	0.6	

LG-100	0.75	with the vacuum degree of the pump
LG150	0.85	
LG200	1	
LGB70	0.45	
LGB110	0.55	
LGB150	0.7	
LGB220A	0.8	
LGB300	0.95	
LGB300A	0.8	

(2) Purge material

① Standard: air filter in the air;

② Using the carburetor: using the device to recycle the cooling of the gas;

③ The use of N_2 or inert gases as the media of purge.

● 充气密封及掺气

在泵运行中，如果迷宫式密封不能满足工艺生产需要时，可以采用充气的方式加强迷宫密封的密封效果，以彻底隔离润滑油腔与泵腔。工业生产中通常采用空气或N_2作为密封介质，在特殊场合可采用惰性气体作为密封介质。密封介质的压力为0.5～ 1.5 Kg/㎠ G；需求量会随泵进气压力与被抽气体的性质不同而改变，可根据实际情况进行调整，充气口位于侧盖的顶部。

掺气主要是通过引入外界气体降低泵内可凝性气体、腐蚀性气体的浓度，降低真空泵内可凝性气体、腐蚀性气体的分压力，避免气体的冷凝与腐蚀等情形发生。真空泵运行时掺气也可以起到**冷却吹扫**的作用。掺气口位于泵体的侧面。

★充气密封与掺气一般选用相同的介质，本型号螺杆真空泵出厂时已将充气密封与泵的掺气的进气管道连接在一起，只留一个输入口，用户在使用时，接通干压缩空气、N_2或惰性气体等即可；客户也可根据实际需要，将掺气管与充气密封管分开接入。

● Iinflatable seal and aeration

In the operation of the pump, if the labyrinth seal fails to meet the process production needs, the sealing effect of the labyrinth seal can be strengthened by aeration mode, and the oil cavity and pump cavity are completely separated. The industrial production is often used as a sealing medium for air or n2, and inert gases can be used as sealing medium for special occasions. The pressure of the sealing medium is 0.5~ 1.5 kg/. The demand will change with the nature of the pump, which can be adjusted according to the actual situation, and the inflatable mouth is located on the top of the side cover.

The gas is mainly used to reduce the concentration of condensate gas and corrosive gas in the pump, reduce the pressure of condensing gas and corrosive gas in vacuum pump, and avoid the condensation and corrosion of gas. The vacuum pump can also be used to cool the effect of the purge. The air inlet is located on the side of the pump.

The gas seal is the same medium as the air gas, the pneumatic pipe channel of the inflatable seal and pump is connected together, and only one input mouth is left, and the user is using the dry compressed air to be compressed air and the inert gas of the air. Customers can also separate the pipe from the inflatable seal tube according to the actual needs.

● 清洁吹扫（重点推荐）

本系列螺杆真空泵采用高精度的螺杆转子与泵体，泵啮合间隙小，在螺杆泵停止工作前，有必要清理泵腔内部。在抽吸腐蚀性/有毒性的气体时，残余的气体会腐蚀转子与泵腔的表面，使泵的啮合间隙变大，导致真空泵性能下降，严重时可使泵失效；在抽吸粘稠物质如树脂等时，停泵后粘稠物质会粘结在转子或泵腔的表面，使泵啮合间隙变小，使泵的运行稳定性下降，严重时可导致真空泵卡死。在抽吸腐蚀性/有毒性的气体或粘稠物质如树脂等时，清洁吹扫尤其重要。

在泵停止工作前，关闭进气端主阀，再将干空气或N_2充入进气口，对泵腔内部进行吹扫，吹扫时间一般可选择10至15分钟。吹扫介质的进气量越大，进气压力越高，吹扫效果越好，进气压力一般不需高于大气压力。

如果普通的清洁吹扫不能有效清除泵腔内的有害物质时，可以使用蒸汽或者其他清洁气体作为吹扫介质进行强化吹扫。强化吹扫完成后，需再进行普通的清洁吹扫，清除泵腔内残余的蒸汽或其他清洁气体。

● Clean and blow (recommended)

This series screw vacuum pump is used to clean the pump cavity inside the pump cavity before the screw pump stops working. In the case of a corrosive/toxic gas, the residual gas will erode the surface of the rotor and the pump cavity, and the air of the pump will be larger, causing the vacuum pump to decline and the pump will fail when it is serious. In the case of the viscous material such as resin, the viscous matter will be stuck in the surface of the rotor or pump cavity, and the pump mesan will be reduced, and the operation stability of the pump will decrease, which can cause the vacuum pump to die. Cleaning is especially important when inhaling corrosive/toxic gases or sticky substances such as resin.

Before the pump stops working, close the main valve of the valve, then fill in the dry air or n2, and blow it inside the pump cavity, and the purge time may be chosen for 10 to 15 minutes. The greater the amount of the purge medium, the higher the pressure of the process, the better the blow, the pressure of the system is not higher than the atmospheric pressure.

If the normal cleaning purge does not effectively remove the harmful material in the pump cavity, the steam or other clean gas can be used to strengthen the purge. After the enhanced purge is completed, the normal cleaning will be cleaned, and the residual steam or other clean gas in the pump cavity is cleared.

●蒸汽吹扫

如果螺杆真空泵长时间未使用而在泵重新启动时，发生嵌顿或者负载过重的现象，请不要强行启动、运转，务必按以下步骤对螺杆泵进行蒸汽吹扫：

＊ 蒸汽吹扫方法(在泵停止运行后)

(1) 处理以下物质后需要蒸汽吹扫 ： 有机物,单体,聚合物，树脂等；

(2) 蒸汽压力 ： 约 1 Kg/cm^2G；

(3) 吹扫过程；

① 关闭吸入阀门,打开排出阀门或消音器/分离器的排出阀门；

② 通过泵吸入路径或者进气端连接阀注入蒸汽 1～10分钟(具体时间视不同的处理物质而定)；

③ 将电机后盖拆下，用手转动电机风扇看是否运转顺畅(注意，不能启动电机)；

④ 当泵运转顺畅时，即可停止蒸汽注入。然后，启动电机，看泵是否正常运转。

（4）如果仍未达到满意效果，可以重复以上步骤2～3次。

● Steam purge

If the screw vacuum pump is not used for a long time and when the pump is restarted, the phenomenon of inplantable or overloaded, please do not force to start and run, be sure to blow the screw pump in the following

* Steam purge method (after the pump stops running)

(1) After processing the following substances, the steam will be cleaned: organic, monomer, polymer, resin, etc.

(2) Steam pressure: about 1 kg/cm2 ;

(3) The process of purge;

① The valve is closed and the discharge valve of the outlet valve or the silencer/separator is opened.

② By pumping the suction path or the "the" () "end" connection valve into the steam 1~ 10 minutes (depending on the different processing material);

③ Remove the rear cover of the motor and use the hand to turn the motor fan to see if it is running smoothly (note that the motor machine) can not be started.

④ The steam injection can be stopped when the pump is running smoothly. Then, start the motor and see if the pump works.

(4) If the satisfactory effect is not reached, the above steps can be repeated three times.

2. 操作程序

2.1 安装要求

● 位置

（1）真空泵需安装在清洁、平坦、坚硬的基础上，基础可以使用混凝土的地基，也可使用金属框架。基础必须具备足够的强度，充分考虑真空泵的重量和基出的承受面积，以免真空泵因基础变形而受力，受力严重时可导致真空泵扭曲变形，破坏真空泵的内部结构，极易使真空泵出现运行故障，甚至导真空泵出现无法修复的损坏。

（2）如果真空泵需在室外使用，则需确认电机及其他部件是否适合于室外使用。

（3）真空泵的安装位置设计时，必须保证留有足够的设备维修、保养空间，以便于设备的定期检查、日常维护保养，以及设备的拆分、重装。

2.Installation requirements

● Position

(1) The vacuum pump should be installed on a clean, flat and hard basis, and the foundation can use the foundation of the concrete and the metal frame can be used. The foundation must have sufficient strength, fully consider the weight of vacuum pump and the area of the base, so as not to be forced by the basic deformation, the pressure can cause the vacuum pump to distort the deformation, destroy the internal structure of the vacuum pump, easily cause the vacuum pump to run the fault, and even the vacuum pump can not repair the damage.

(2) If the vacuum pump needs to be used outside, it is necessary to confirm whether the motor and its parts are suitable for outdoor use.

(3) When designing the position of the vacuum pump, it is necessary to ensure that sufficient equipment maintenance and maintenance space is provided for regular inspection of the equipment, daily maintenance, and the separation and reloading of the equipment.

（4）如果真空泵的安装位置附近存在振动源，则必须设计隔振装置，如开设隔振沟等。

（5）基础的四周应留有5到10厘米的槽沟，防止放油或放水时沾污工场。

●安装

（1）本泵在使用前须将泵上的一切尘污全部清除，泵房必须经常保持清洁干净，环境温度应在5～40℃范围内。

（2）真空泵安装时，必须使用水平尺对基础进水平校准，水平误差应小于0.5毫米/米。基础误差过大时，可以使垫铁辅助校准。使用垫铁时，基础受力面积减少，需重新校合基础的强度。

（3）如果与真空泵连接的管道有振动时，可以使用波纹管隔断振动的传播。

(4) If there is a vibration source near the installation of the vacuum pump, the vibration isolation device must be designed, such as opening the vibration channel.

(5) On the basis of the base, you should leave five to 10 centimeters of gutter to prevent the contamination of oil or water.

● Installation

(1) The pump must be cleaned up before use, and the pump room must be kept clean and clean, and the ambient temperature should be within 5~ 40c.

(2) When the vacuum pump is installed, the basic water level calibration must be used with the horizontal rule, and the horizontal error should be less than 0.5 mm per meter. When the base error is too large, the cushion iron auxiliary calibration can be made. When using pad iron, the base force area is reduced, and the strength of the foundation is required.

(3) If the pipe connected to the vacuum pump vibrates, the propagation of the vibration of the corrugated tube can be used.

2.2 管道联接

●主管道

（1）清洗吸入和排出管内部，以除去铁锈、灰尘及异物等，并在螺杆真空泵的运行初期，在吸入口处配过滤网，并必须定期检查，确保滤网留存足够的通道面积，以免滤网被吸入泵腔，造成真空泵损坏。真空泵运行一段时间后，在确认管道内再无较大硬质异物后，可去除滤网。

（2）建议在吸入口和排出口处使用软连接，如波纹管、橡胶管等，这样可以减少对泵体不必要的负载。

（3）务必在吸入口或排出口处安装止逆阀使得泵不会发生反向抽吸。如果客户无法安装止逆阀的话，至少也需要安装一个闸门阀，并保证在停止泵前将其关闭。

（4）若在泵排出管会有冷凝气体时，置一个回收罐在泵下，这样冷凝气体就可以在泵工作中收集并通过排泄阀排出。

（5）收集冷凝气体需注意，冷凝器前端需要安装排泄阀。

（6）进气管路应该非常严密，即使微小的泄漏都会影响真空度，进气管路原则上尽可能短，少用接头和弯头，与泵联接的管道不应少于泵的口径。

2.2 Connection

● Main pipe

(1) Cleaning and discharge of rust, dust and foreign bodies, and in the early days of the screw vacuum pump, and the network of the suction mouth, and must be inspected regularly to ensure that the filter is retained enough

area to prevent the filter from being sucked into the pump cavity, causing the vacuum pump to be damaged. After the vacuum pump runs for a period of time, the filter can be removed after the confirmation of the pipeline has no larger rigid foreign object.

(2) It is recommended to use soft connections in the inhalation and outlet, such as corrugated pipes and rubber tubes, which can reduce the load of unnecessary pumps.

(3) It is important to install the check valve at the suction or outlet, so that the pump does not take a reverse suction. If the customer can't install the reverse valve, at least a gate valve is installed, and it is guaranteed to be closed before the pump is stopped.

(4) If a pump is used to pump down the pump, a recovery can is pumped under the pump, which can be collected in the pump and discharged through the discharge valve.

(5) to collect the condensate gas attention, the condenser front end needs to install the discharge valve.

(6) The passage of the airway should be very tight, and even if the small leakage affects the vacuum, the inlet channel is as short as possible in principle, with no joint and elbow, and the pipe that is connected to the pump should not be less than the diameter of the pump.

●冷却水管道

对于本系列螺杆真空泵，冷却系统采用直接水冷的方式冷却端盖甲、侧盖和泵体。真空泵内部的冷却水管路已经连接，客户只需接通进、出水管路即可使用，注意冷却水需遵循**低位进、高位**出的原则；冷却水进水管路上应装阀门，可以调节水量。

螺杆真空泵运行时，转子的温度最高可达200℃，运行时适当地减少冷却水量，使泵体保持较高的温度（比如80~90℃），有利于减少泵体与转子之间的温差，增大泵体与转子之间的间隙，从而提高真空泵的运行稳定性。另外，在抽含可凝性气体的物质时，可适当减少冷却水量，使真空泵保持较高的温度，可以有效减少气体在泵腔内凝结。

2.3 开机准备

（1）清除泵中的尘土，管子要确保清洁彻底，不能有焊接渣残留在内。

（2）检查所有吸入、排出管道以及冷却水管道口是否正确上紧，所有管子是否安装无误。

（3）加油至油表红点处。如果油过少，齿轮和轴承会咬死；如果油太多，温度会过度升高，导致齿轮噪音或者对其他部件的影响。所以，请注意将油位长期保持在油表红点处。

（4）请将冷却水量按照表1.3的要求配置。

● Cooling water pipe

For this series of screw vacuum pumps, the cooling system adopts direct water cooling to cool the end cover and the side cover and the pump. The cooling water line inside the vacuum pump is connected, the customer only needs to connect in, the outlet line can be used, notice the cooling water needs to follow the low entry, the high point out principle; The water intake pipe should be fitted with the valve to adjust the water.

When the pump is run, the rotor temperature can reach 200℃, and the cooling water quantity is properly reduced, and the temperature of the pump is maintained by the high temperature (such as 80~ 90℃), which is helpful to reduce the temperature difference between the pump and the rotor, and increase the gap between the pump and the rotor, thereby increasing the stability of the vacuum pump. In addition, when the substance of the condensing gas is drawn, the water content can be reduced properly, and the temperature of the vacuum pump is maintained, which can effectively reduce the condensation of the gas in the pump cavity.

2.3 Boot preparation

(1) Remove dust from the pump, and ensure that the pipe is clean and thoroughly, and no welding slag remains.

(2) Check all the suction, discharge pipes and cooling water pipes to be properly tightened, and all the pipes are installed.

(3) Go to the oil table red. If oil is too low, gears and bearings will bite. If you have too much oil, the temperature will go up,

The effect of the gear noise or other parts. Therefore, please note that the oil level is maintained in the oil table in the red spot for a long time.

(4) please configure the cooling water quantity according to the requirements of table 1.3.

2.4 运转

(1) 打开吸入阀，在无负载状况下点动真空泵，以检查转子转向（观察电机风扇转向）。

(2) 让泵在无负载情况下运转20-30分钟，以检查是否有非正常的振动或发热。如果有的话，请将机器停转并寻找原因。在大部分情况下，导致的原因通常是安装不正确或位置没有摆正。还有时候是因为润滑剂缺少。

(3) 泵在正常负载情况下运转2 - 3小时后，请检查一下各个部分的温度和振动情况。

(4) 运转过程中，请注意观察安培计。如果发现任何不正常状况，请立即停止泵工作并查找原因。常见的原因是转子和转子之间的碰撞或者转子外圈与泵体内面的接触。所有我公司出品的泵类产品均已通过该方面的测试。尽管如此，泵的状态仍需要严密关注，毕竟在现场，泵又经过了重新安装。

＊ 运转过程中的注意事项 ＊

(1) 检查轴承和润滑剂的温度以及冷却水。

(2) 让泵保持在设计范围内工作。

2.4 Operation

(1) Open the suction valve and move the vacuum pump at the non-load condition to check the rotor steering (see the motor fan to the).

(2) To allow the pump to run for 30 minutes without load, to check if there is any abnormal vibration or heat. If there is a word, please stop the machine and find out why. In most cases, the reason is that it is usually wrong or misplaced. There is also a lack of lubricants.

(3) When the pump is operating for 23-3 hours, please check the temperature and vibration of each part.

(4) In the course of operation, please pay attention to the ammeter. If you find any errors, please stop the pump and find out why. The common cause is the collision between the rotor and the rotor or the outer ring of the rotor and the pump body. All the pump products produced by our company have been tested in this area. Nevertheless, the state of the pump still needs to be closely watched, after all, and the pump is reinstalled.

* Notice in the operation of the operation *

(1) Check the temperature and cooling water of bearings and lubricants.

(2) Keep the pump in the design range.

2.5 停止

(1) 将进气端主阀关闭。

(2) 如果吸入过腐蚀性的气体，请务必在停泵前进行清洁吹扫工作。

(3) 通过关闭电机将泵关闭。

(4) 再将冷却水系统关闭。

注：如果存在严寒结冰现象，打开排泄阀将泵内的冷却水除尽，以免真空泵冻裂。

2.6 润滑

润滑油请使用质量好的高等级的石油产品包括抗氧化剂，防腐蚀剂，超高压添加剂等成分（不要使用含有水分，硫酸树脂或焦油成分的润滑剂），出厂标配100#真空油。

透平油(ISO 46) 在市场上能够获得，能够广泛满足该类泵产品的需求。

以下品牌为推荐轴承和齿轮使用的润滑油种类。

・润滑剂 : BP, Energol THHT 46 / THB46
Regal, R & O 46
Shel , Turbo T 46
Mobil, DTE oil medium or, equivalent.

- 特殊使用: Fomblin, 25/6

・油 脂 : Almasol 1262
G 40M (SHINETSU)
JFE 552 (KLUBER)
Arcanol L74V (FAG)

- 特殊使用: RT-15 (Fomblin)
Barrierta L55/2 (KLUBER)

2.5 Stop

(1) Turn off the main valve.

(2) If the corrosive gas is inhaled, be sure to clean and purge the work at the stop pump.

(3) Close the pump by closing the motor.

(4) Turn down the cooling water system.

Note: if there is a cold freeze, open the discharge valve and remove the cooling water in the pump so as not to freeze the vacuum pump.

2.6 lubrication

Lubricating oil is used for high quality oil products including antioxidants, anti-corrosive agents, ultra-high voltage additives, etc. (don't use lubricants that contain water, sulfuric acid or tar components, and 100 # vacuum oil.

The oil (iso 46) can be obtained in the market and can be widely met with the demand of such pump products.

The following brands recommend the type of lubricants used for bearings and gears.

The lubricant: bp.energol THHT 46 / thb46
Regal, r & o 46
Shell, turbo t 46
Mobiles, dte oil medium, equivalent.

- special use: fomblin.25/6

Oil fat: almasol 1262
G 40m (shinetsue)
JFE 552 (klubers)
Arcanol l74v (

- special use: RT-15 (Fomblin)
Barrierta L55/2 (KLUBER)

3. 维护和检查

3.1 概述

(1) 在运转过程中，温度会根据压缩比的上身而上升。如果温度上升是局部的，而且外表面很烫，那就是非正常现象。可能的原因是，冷却水供给出现问题或者是冷却系统未开，转子与泵体摩擦或者是泵内进入了异物。此时，必须立即将泵停止并检查原因。在某些情况下，转子和泵体可能经过长期运转而腐蚀，从而导致这些部件之间的间隙变大，排出气体返回吸入口，抽吸能力较原设计能力下降。这种情况下，抽气量会明显下降，需要停止运转，测量误差，考虑采取修正措施。

(2) 通过对泵体、轴承的温度、振动和噪音的常规检查，可以立即发现异常现象。因此，强烈建议每日对泵例行检查。

(3) 转子和转子之间的碰撞或者转子外圈与泵体内面的接触能够通过在外表面上放置听诊装置听到。所以，注意泵发出的声音也很重要。

(4) 在寒冷地区的冬季，泵停止后，将冷却水排出，以免水冻结后对泵体、侧盖造成损伤。

3.Maintenance and inspection

3.1 Overview

(1) In the course of operation, the temperature increases according to the compression ratio. If the temperature rise is local and the surface is very hot, that is the abnormal phenomenon. The possible reason is that the cooling water supply problem or the cooling system is not open, the rotor is friction with the pump or the pump enters the foreign body. At this point, the pump must be stopped and the reason is checked immediately. In some cases, the rotor and the pump may be corroded by the long run, resulting in a large gap between the parts, the discharge of the gas returning to the inhalation, and the decline in the ability to pump. In this case, the amount of extraction will be significantly reduced, and the operation will need to be stopped, the measurement error is measured, and the corrective measures will be considered.

(2) The abnormal phenomena can be detected immediately by regular inspection of the temperature, vibration and noise of the pump body, bearing. Therefore, it is strongly recommended that the pump routine be checked daily.

(3) The collision between the rotor and the rotor or the outer ring of the rotor and the pump body can be heard on the outer surface. Therefore, it is also important to note the sound of the pump.

(4) In the winter of the cold region, when the pump stops, the cooling water is discharged, so as not to be damaged by the water freeze.

3.2 周期性检查内容

- 日常检查项目
  - ① 检查油表：过多或过少的润滑剂都会损伤齿轮和轴承。
  - ② 检查冷却水量是否足够。
  - ③ 检查后盖和前端盖的温度。使用适当的温度计来测量，使用表面温度计将会更加方便。
  - ④ 检查吸入和排出端压力。测量压力以保证泵在规格允许的范围内工作。
  - ⑤ 检查电机负载。注意电机负载的增加往往提示着非正常的状态。
- 月度检查项目

① 检查润滑油的颜色(如果油的颜色变深，请更换润滑油)。

② 检查油量，如果油大量减少，请检查泵内部的密封结构。

● 季度检查项目

① 检查冷却水流量和颜色。如果发现不正常，请清洗冷却水夹层和水管。

② 检查后端盖的油脂，必要时增加。

● 半年检查项目

① 检查泵连接的管路。

② 更换前端盖润滑剂。

③ 更换后端盖油脂。

● 年度检查项目

① 检查真空泵内部动密封。

② 检查内部转子与泵体。将进气端的吸入管卸下，以查看内部转子和泵体的状况。

③ 检查齿轮。将前端盖打开以检查齿轮的状况。

3.2 Periodic check content

● Daily inspection project

① Check the oil meter: too much or less lubricants can damage gears and bearings.

② Check to see if the cooling water is sufficient.

③ Check the temperature of the rear cover and the front cover. Using a proper thermometer to measure, using a thermometer thermometer will be more convenient.

④ Check inhalation and discharge pressure. Measure pressure to ensure that the pump works within the scope of the specification.

⑤ Check the motor load. Note that the increase in motor loads often indicates abnormal state.

● Monthly inspection project

① Check the color of the lubricating oil (if the color of the oil becomes darker, please replace the lubricating oil).

② If oil is reduced, check the sealing structure inside the pump.

Quarterly inspection project

① Check the water flow and color. If you find improper, please clean the cooling water layer and water pipe.

② Check the grease of the rear cover and increase when necessary.

● Semi-annual inspection project

① Check the piping of the pump.

② Replace the front cover lubricant.

③ The rear end of the grease.

Annual inspection project

① Check the internal motion seal of the vacuum pump.

② Check internal rotor and pump body. Remove the suction pipe of the machine to view the internal rotor and pump body.

③ Check the gear. Open the front cover to check the gear situation.

干式螺杆真空泵维修检测清单

项目	检测要点	每日	每月	3~6月	每年
1.电机电流	是否有变化？ 是否符合标准要求？	☺ ☺			
2.螺杆旋转	旋转是否顺畅及正确？	☺			
3.进、排气压力	压力是否在正常范围内？	☺			
4.噪音及振动	是否有不正常的噪音及振动？	☺			
5.温度	泵体外表温度是否在标准范围内？ 轴承处温度是否在标准范围内？	☺ ☺			
6.前端盖油位	油位是否在正常水平？	☺			
7.前端盖油质	前端盖油是否乳化或污染？	☺			
8.漏油	是否存在漏油现象？	☺			
9.润滑剂更换	前、后端盖的润滑剂须定期更换。 后端盖的润滑脂须定期检查。		☺	☺	
10.防护冷却水量及压力	冷却水量是否符合标准？ 水压是否在标准范围内？	☺ ☺			
11.进、排气管道路	是否有污垢？			☺	
12.清洁吹扫	每次停泵前进行清洁吹扫	☺			
13.检测泵腔及螺杆	是否存在锈迹或腐蚀现象？ 表面是否有裂纹？				☺ ☺
14.泵的易损件	必要时进行更换				

3.3拆卸

● 拆卸的注意事项

① 任何拆卸的工作必须由专业的维修技术人员担任，维修人员必须熟悉真空泵相关的安全守则，必须熟悉螺杆真空泵的拆装步骤。

② 真空泵完全停转前不得进行移动、拆卸等操作，并需确认已切断电源。

③ 在抽除有毒、腐蚀性等对人体有害的气体时，拆卸前，请先使用N_2对真空泵进行全面清扫，务必使残留气体尽数排出。

④ 清注意，运行时真空泵零部件温度很高，请在泵彻底冷却后再行拆装，以免烫伤。

⑤ 注意不要损伤连接处的包装。在所有的链接和吻合处标注模缝线。

⑥ 卸下的部件需要防尘处理。特别主要保护好密封零部件，齿轮以及轴承。

● 拆卸步骤

1 将泵上的所有附属物移除。

2 打开泵体下方的排泄阀，排出冷却水。

3 打开前端盖上的排油塞将油排出。

4 松开电机连接螺母，拆下电机。

5 松开联轴器上的螺钉，取下甩油板、和联轴器。

6 松开螺母，取下端盖甲。

7 松开齿轮胀圈螺钉，将主、从动齿轮卸下。

Dry screw pump maintenance test list

The project	detects the point	daily	Mouth	3~6M	YEAR
1. Motor current	Does it change? Is it consistent with standard requirements?	☺ ☺			
2. Screw rotation	Is the rotation smooth and correct?	☺			
3. Inlet, exhaust pressure	Is the pressure normal?	☺			
4. Noise and vibration	Is there any noise and vibration?	☺			
5. Temperature	Is the temperature of the pump in the external table in the standard range? Is the temperature of the bearing temperature within the standard range?	☺ ☺			
6. Front cover oil level	Is the oil level normal?	☺			
7. Front cover oil quality	Is the front-end cover oil emulsified or polluted?	☺			
8. Oil leak	Is there a leak?	☺			
9. Lubricant replacement	The lubricant of the front and rear end should be replaced regularly. The grease of the rear end should be inspected regularly.		☺	☺	
10. Protective cooling water quantity and pressure	Does cooling water meet the standard? Is the water pressure within the standard range?	☺ ☺			
11. Inlet and exhaust pipe channel	Is there any dirt?			☺	
12. Clean purge	Clean and purge every time the pump is stopped	☺			
13. Test pump cavity and screw	Is there rust or corrosion? Is there a crack on the surface?				☺ ☺
14. Pump's vulnerability	Change when necessary				

3.3 Removal

● Remove the notice

① Any disassembled work must be served by professional maintenance technicians, and the maintenance personnel must be familiar with the safety rules of the vacuum pump, and must be familiar with the removal and removal of the screw vacuum pump.

② The vacuum pump shall not be used for moving and disassembling before the vacuum pump is completely shut down, and the power supply is confirmed.

③ Before removing toxic, corrosive gases and other harmful gases, please use n2 to complete the vacuum pump for total cleaning first, so that the residual gas shall be discharged.

④ The vacuum pump parts temperature is very high, please remove it after the pump is completely cooled, so as not to burn.

⑤ Note not to damage the packing of the connection. Mark the pattern seam in the links and anastomoses.

⑥ The components that are unloaded require anti-dust treatment. Special protection of sealing parts, gears and bearings.

● Tear down step

1 Remove all the appendage of the pump.

2 Open the discharge valve below the pump and discharge the cooling water.

3 Open the oil drain on the front lid and drain the oil.

4 Loosen the motor connection nut and remove the motor.

5 Loosen the screw on the coupling and remove the oil plate and the coupling.

6 Loosen the nut and remove the lower top.

7 Release the gear expansion screw, remove the main and the gear gear.

8 接着先拆进气口侧的零部件。松开螺钉，将端盖乙盖板卸下

9 松开螺栓，将轴承压板卸下。

10 松开端盖乙上的螺栓，使用M12的顶杆将端盖乙从泵体上卸下。

11 将波纹弹簧、深沟球轴承、干式密封与活塞环套乙从端盖乙中卸下。

12 接下来再拆排气口侧的零部件。松开轴承压盖的螺钉，卸下轴承压盖。

13 松开圆螺母上紧定螺钉，卸下从、从动轴上的圆螺母，注意紧定螺孔内的铜垫。

接下来是拆侧盖内的零件，这些零件是用于确定螺杆固定端端面间隙，因此在拆卸时须作好标记，以便于重装时原样复位，避免重新调整固定端间隙。

14 使用的螺栓将轴承座带轴承一起从侧盖上卸下，再将轴承从轴承座内卸下，注意轴承座内是否存在调整垫圈。拆卸时注意将轴承座与垫圈作好标记。

15 使用的顶杆将侧盖从泵体上卸下，再从侧盖中取出干式密封，注意是否存在调节垫圈，并在垫圈上作好标记。

16 卸下主、从动转子上的活塞环套甲、轴套，并作好标记。

17 轻轻的从泵体内推出主、从动转子，并将它们用尼龙绳悬挂起来。

18 将水套盲板从泵体上卸下（用于清洗泵体水隔层，可不拆）。

使用高品质的清洗剂清洗部件并按照工厂制定的范围更新旧的或者已经损坏的部件。每次组装建议更换新的轴承，密封，垫圈和O型圈。

8 Remove the parts of the inlet side. Loosen the screw and unload the end cap

9 Release the bolt and remove the bearing board.

10 Loosen the bolt on the end of the end cap and remove the end cap from the pump by using the M12 top rod.

11 The corrugated spring, the deep groove ball bearing, dry seal and piston ring set b are removed from the end cap.
12. Then remove the parts of the exhaust side. Loosen the screw of the bearing gland and remove the bearing gland.
13. Release the nut to tighten the screw, remove the round nut from the axis, and pay attention to the copper mat inside the screw hole.
The next step is to remove the parts in the side cover, which is used to determine the fixed end of the screw, so it must be marked in the disassembly so that the reload is reinstalled at the time of the reloading, and avoids the adjustment of the fixed end clearance.
14. The bolt is used to remove the bearing bearing bearing and remove the bearing from the side cover, and then remove the bearing from the bearing seat and notice if there is a adjusting gasket in the bearing seat. Note the mark of the shaft bearing and the gasket when disassembly.
15. The top rod is removed from the pump, and the dry seal is removed from the side cover, and notice if there is a adjusting washer and marks on the washer.
16. Remove the piston ring of the rotor on the rotor and make a good mark.
17. Gently launch the main, slave rotor, and hang them on the nylon cord.
18 Remove the water cover plate from the pump.

Use a high-quality cleaning agent and update the old or damaged parts according to the scope of the factory. New bearings, seals, gasket and o rings are recommended for each assembly proposal.

3.4组装

- 组装注意事项
  ① 组装时请注意检查是否有旧的或者已经损坏的部件。特别是连接处或者吻合处的损伤将会严重影响组装后的效果。所以，细致的检查连接处或吻合处的磨损或损伤非常重要，如有发现，请更换或者维修。
  ② 用轻油清洁轴承，然后上润滑油。任何时候处理轴承时，请使用干净的工具并洗手。
  ③ 使用软布和清洁剂对吻合部件除尘，并涂上油。对于紧实吻合部件建议使用二硫化钼，因为这些部件一旦锈蚀将很难卸下。组装会比拆卸要困难一些。（对于齿轮的锥形部件，在安装前用软布和清洁剂彻底清洁表面）。
  ④ 使用跟旧包装厚度一样的新包装来包装。
- 组装步骤
  1. 将后盖板用螺钉固定在端盖乙上，再将后盖板置于工作台或合适的管架。
  2. 在端盖乙中垂直插入主、从动转子，注意主、从动转子的轴伸较短的的一端朝下，然后在转子的上伸出端装上0形密封圈。
  3. 组装从泵齿轮端开始（即前端和排气端），原零部件的配对关系。
  4. 按原配对样式将活塞环套甲、干式密封与轴套装入侧盖中，安装时尽量使活塞环与环套保持同心。再根据正确的配对关系，使转子伸出轴通过相应的活塞环套甲和轴套的内孔，装配时注意压住轴套，防止轴套被带出干式密封，再将侧盖置于主、从动转子上，此时，侧盖的下端面应与螺杆转子的上端面接触。
  5. 再将调节垫圈按原配对关系安装在转子轴上。
  6. 接着将轴承座按原配对关系安装在转子轴上。然后在轴承座中装上双列向心球轴承。
  7. 使用圆螺母将轴承固定在转子轴上，注意圆螺母带锥面的一端朝里。再将圆螺母上的紧定螺钉拧紧，注意在紧定螺钉的前面必须安放铜垫，否则会损坏转子轴上的螺纹。
  8. 装上轴承压盖，并用螺钉固定在侧盖上。
  9. 用厚度尺(= 测隙规)，检查转子端面与侧盖之间的间隙是否与原先相同。

10. 提起组装好的部件，将其与端盖乙分离，以组装泵体。把组装好的部分上下颠倒过来，使前端部分放在操作台上，后端部分朝上的垂直放置。安放时可使用等高木质垫块，并以侧盖端面为支撑面，安放时注意保护侧盖端面。

3.4 Assembly

● Assembly considerations

① When assembling, please note whether there are old or damaged parts. Especially the damage of the connection or anastomosis will seriously affect the effect of assembly. Therefore, it is very important to examine the wear or damage of the connection or anastomoses, if found, please replace or repair.

②Clean the bearing with light oil and then grease. When handling bearings at any time, use clean tools and wash your hands.

③ Use soft cloth and cleaner to dust the anastomosed parts and apply oil. The use of disulfide is recommended for the compact anastomosis, because these parts will be difficult to remove once the rust is eroded. Assembly will be more difficult than dismantling. (for the tapered parts of the gear, use soft cloth and cleanser before installation).

④ Use the new packaging as thick as the old packing thickness.

● Assembly step

1. The rear plate is fixed on the end of the end cap and the rear plate is placed on the top of the end cap, and the back plate is placed on the workbench or the appropriate pipe frame.

2. In the end of the end of the end of the shell, the rotor is inserted vertically into the main, the slave rotor, the main, the shorter end of the shaft of the rotor rotor, and then the upper ring of the rotor on the upper end of the rotor.

3. The assembly is assembled from the pump gear end (that is, the front-end and exhaust end), the matching relationship of the original parts.

4. The piston ring and the ring and the ring are the same as the ring and the ring, which is the same as the type of the piston ring, the dry seal and the shaft sleeve. According to the correct matching relationship, the shaft is used to prevent the shaft of the shaft to be sealed by the shaft, and the side cover is placed on the main and the rotor, and the lower end of the side cover should be exposed to the upper side of the screw rotor.

5. The balance washer is installed on the rotor shaft.

6. The bearing seat is then mounted on the rotor axis by the original matching. Then put the double column to the core ball in the bearing seat.

7. Use a round nut to attach the bearing to the rotor shaft and notice the round nut with the cone on the end of the line. Then tighten the fixed screw on the round nut and note that the copper mat must be placed in front of the screw, otherwise the thread on the rotor axis will be damaged.

8. Load the bearing cover and fasten it to the side cover.

9. Use the thickness of the thickness rule (the) to check whether the gap between the rotor and the side cover is the same as the original.

10. Bring up the assembled parts and separate it from the end of the valve to assemble the pump. Turn the assembly part upside down and put the front end part on the console, and the rear end of the end is placed vertically. When placed, the high wood block is used, and the side cover is supported, and the side cover is protected.

11. 在侧盖向上的端面的各密封槽内装入0形密封圈，再将泵体按正确位置装于侧盖上，确认泵体下端面与侧盖端面吻合。再将挡尘圈装在轴上。
12. 将端盖乙的密封槽内装入0形圈，再将其装于泵体上，并用螺栓固定。
13. 将活塞环套乙、干式密封与轴套装在端盖乙中。
14. 在端盖乙的轴承安装孔的0形圈槽内安放0形圈，再进行轴承的安装，注意轴承朝里的一面须带密封圈，并确认轴承已经上好润滑油脂；再装上轴承压板，并用螺栓固定。注意固定时必须配弹簧垫圈。
15. 将波形弹簧安放在轴承安装孔内，并在端盖乙外端面的密封槽内装入0形密封圈，再将后盖板装上，并用螺栓固定。
16. 把组装好的部件提起，并将其水平放置于操作台上。
17. 将从动齿轮安装到轴上，先适当拧紧齿轮胀圈的螺钉，固定从动齿轮，再将百分表座安装在侧盖上，测从动齿轮端面跳动量，跳动量过大时，可通过轻击齿轮端面高点的方法来调整，当端面跳动量小于0.05㎜时，再将螺钉拧紧，最终固定从动齿轮。接着将主动齿轮安装到轴上，适当带紧齿轮胀圈的螺钉，使主动转子与齿轮能相对转动，再通过在进气口将适当厚度的厚度尺(= 测隙规)插入主、从动转子之间的啮合缝隙中，并稍微转动从动齿轮，让厚度尺固定住，适当拧紧主动齿轮胀圈的螺钉。取出厚度尺，然后测量转子之间的啮合间隙，间隙均匀后，再调整齿轮端面跳动量，当跳动量小于0.05㎜时，将螺钉拧紧，最终固定主动齿轮。**本步骤是泵安装的关键步骤**。
18. 将双头螺柱拧在泵体上，在侧盖外端面的密封槽内装入0形密封圈，将端盖甲装在侧盖上，用弹簧垫圈与螺母将其固定。
19. 将联轴器与甩油板安装到主动齿轮上，用螺钉固定。
20. 在端盖甲的外端面上安放密封垫，将电机安装在端盖甲上，并用弹簧垫圈与螺母固定。
21. 从前盖板顶端的油进口处加入润滑油。油量需要达到油镜中线稍上即可。(每款产品的油量见规格表1.3)。
22. 将水套盲板固定在泵体上。
23. 安装各种附件。真空泵组装完成。

11. The oiled seal ring is loaded inside the sealed slot of the upper surface of the side cover, and the pump body is placed on the side cover in the correct position, confirming that the bottom side of the pump is consistent with the side cover. Put the dust ring on the shaft.
12. load the o-ring in the sealing slot of the end lid b and put it on the pump and fixed with the bolt.
13. The piston ring b, dry seal and sleeve are installed in the end cap.
14. In the o ring slot of the bearing hole of the end of the end of the end of the socket, the ring is placed in the slot of the hole, and the installation of the bearing is carried out, and the sealing ring of the bearing in the bearing is required, and the bearings are made of grease. Put on the bearing press and fixed with bolts. Note that the spring washer must be fitted when fixed.
15. The corrugated spring is placed in the shaft of the bearing, and the open ring is loaded into the o-shaped seal in the sealing slot of the outer end of the end lid, and then the rear cover is mounted and the bolt is fixed.
16. Bring up the assembly and place it on the console.
17. The screw is installed on the shaft, fixed the gear gear, and the percentage of the dial seat is installed on the side cover, and the speed of the gear face is measured, and the method of reducing the top height of the gear is adjusted by the way of the gear, and the screw is tightened and the gear is fixed when the face of the gear is less than 0.05mm. Then, the screws of the gear can be fastened to the gear, allowing the active rotor and the gear to be relative, and then insert

the thickness of the proper thickness of the proper thickness into the main, the intermesong slot between the rotor rotor, and a little rotation of the gear, which is fixed in the thickness of the thickness rod, and properly tighten the screw of the active gear expansion circle. The thickness of the line is taken out, and then the intermesh gap between the rotor is measured, and the time of the gear is adjusted when the gap is uniform, and when the beating quantity is less than 0.05mm, the screw is tightened and the active gear is fixed. This step is a key step in the installation of the pump.

18. In the pump body, the double head screw is unscrewed on the pump body, and the upper armor is mounted on the side cover in the sealing slot of the outer end of the side cover, and the spring washer and nut are fixed.

19 . Install the coupling and oil dump plate on the active gear and fix it with screws.

20. The sealing pad is placed on the outer end of the top armor, which is installed on the end cover and fixed with a spring washer and nut.

21. Oil imports at the top of the top plate are added to the oil. The amount of oil needs to reach the center of the oil mirror. (The amount of oil per product is in the specification table 1.3).

22. Fixed the water sleeve blind plate on the pump body.

23. Install various accessories. Vacuum pump assembly is completed.

3.5质保

(1) 质保期限为发货后一年，任何在此期限内由于卖方原因导致的产品缺陷，做工及装配的问题都将由卖方承担处理。

(2) 以下情况将不列为免费质保享受到范围。

① 非卖方能够控制的的因素导致的问题：如自然灾害、火灾等。

② 由于特殊气候导致的产品问题，如气温过高，盐度过大等。

③ 由于使用者未经卖方许可的任意拆改导致的故障。

④ 由于操作、维护不当导致的故障。

⑤ 干式真空泵在操作环境、条件恶劣的情况下运转，手册强调的日常维护检查非常重要。并且，尽管日常例行检查，一定间隔时间如三个月，六个月的全面检查依然十分必要。在特殊操作条件下还需要特别的维护，如半导体行业操作或化工行业需要处理腐蚀性/有毒气体或固体时。

3.5 Warranty

(1) After the delivery of the warranty period, any problem arising from the product defects caused by the seller's cause in this period shall be handled by the seller.

(2) The following circumstances shall not be listed as a free warranty.

① The problem of non-seller can control factors: natural disasters, fires, etc.

② The problem of product due to special climate, such as high temperature, excessive salinity, etc.

③ The failure of the user without the permission of the seller.

④ The failure caused by improper operation and maintenance.

⑤ It is very important to work in the operation of the dry vacuum pump in the operation environment and the condition of the condition. And, despite routine checks, a total interval of time is three months, and a six-month comprehensive examination is still necessary. Special maintenance is required under special operating conditions, such as semiconductor industry operations or chemical industries that need to deal with corrosive/toxic gases or solids.

4. 常见问题和解决办法

问　题	原　　因	解 决 方 法
过热	・ 前端盖润滑油过多 ・ 泵进气温度过高 ・ 压差过大 ・ 转子与泵体摩擦 ・ 冷却水量不足	・ 检查油量 ・ 进气增加冷凝器冷却被抽气体 ・ 排空并清洗排气管道 ・ 查找摩擦的原因 ・ 检查冷却水压力和流量调节阀，清洗冷却水管路
进气量不足	・ 过滤器阻塞 ・ 间隙过大 ・ 进气管路中过滤器阻塞	・ 清洗或更换过滤器 ・ 检查间隙 ・ 清洗或更换过滤器
电机超负荷	・ 机器内进入异物 ・ 管道的压力损失增加 ・ 排气口阻塞 ・ 转子之间摩擦 ・ 转子与泵体之间摩擦	・ 检查泵腔内情况 ・ 检查进出口的压力 ・ 排空并清洗排气管道 ・ 调整齿轮来调整转子之间的间隙 ・ 查找摩擦的原因
爆震音	・ 转子间隙不均匀 ・ 零件组装不正确 ・ 非正常压力增高 ・ 齿轮非正常磨损	・ 重新调整至合适的位置 ・ 重新组装 ・ 寻找压力增高的原因 ・ 更换齿轮
轴承/齿轮损坏、轴破损	・ 润滑油使用不当 ・ 润滑油缺少 ・ 负载过大	・ 更换润滑油 ・ 添加润滑油 ・ 更换轴承或齿轮，并调整负载

*如果通过上述提示操作故障仍然不能够排除，那么问题可能是由于泵的操作条件导致的。这种情况下，请联系我们并注明以下信息：

1. 型号名称和数量，系列名称，应用领域等。
2. 管道的相关信息(阀门,过滤器,弯头数等)。

4. A common solution to a problem

Question	Primary cause	Solution method
Overheat	・ Too much lubricating oil・ ・ The pump is too hot ・ Overpressure ・ Rotor and pump friction ・ Insufficient cooling water	・ Check oil ・ The condenser increases the condenser to cool the gas ・ Empty and clean the exhaust pipe ・ Find the cause of friction ・ Check the cooling water pressure and flow control valve and clean the cooling water line
Insufficient intake	・ Filter blocking ・ Overclearance ・ The filter is blocked in the intake line	・ Wash or replace filters ・ Inspection clearance ・ Wash or replace the filter

Motor overload	· The machine enters the foreign body · Increased pressure loss in the pipeline · Exhaust block · Rotor friction · Friction between the rotor and the pump	· Check the internal condition of the pump · Check the pressure of import and export · Empty and clean the exhaust pipe · Adjust the gear to adjust the gap between the rotor · Find the cause of friction
Knock	· The rotor clearance is uneven · The parts are not assembled · Abnormal pressure increases · Abnormal gear wear	· Reorient to the right place · Reassemble · Find the cause of higher stress · Change gear
Bearing/gear damage, shaft breakage	· Improper lubricating oil · Oil shortage · Overload	· Change oil · Adding oil · Replace bearing or gear and adjust the load

* If the error of the prompt operation is still not enough, the problem may be due to the operating conditions of the pump. In this case, please contact us and specify the following information:The name and number of the model, the series name, the application area, etc.The relevant information of the pipe (valve, filter, bend number, etc.).

5. 真空泵日常操作检查清单

	检　查　项　目	情　况
操作前检查	打开冷却水供给阀门，水流正常吗 ？	
	关闭泵进气管路上的阀门。	
	打开泵排气管路上的阀门。	
	检查润滑油的颜色和量，是否正常？	
	在吸气口打开之前开启泵运转一阵。	
操作中检查	在完全真空情况下检查泵真空度．正常吗?	
	在完全真空下检查电的情况(电压和电流)。合理吗?	
	有异常的声响吗 ？	
	检查泵各部位温度，正常吗?	
	检查润滑油的颜色和量，是否正常？	
停止后检查	关闭吸气管路上的阀门后，继续空运转泵几分钟。	
	如果吸入了异物，需要用清洗剂清洗。	
	关闭冷却水，如果泵长时间不运转，将泵内的冷却水排出。	
	确保进气管路和排气管路上的阀门关闭。	
	确保电源关闭。	

5. Check list of daily operation of vacuum pump

	Check item	mood
Preoperation inspection	Is it normal to open the cooling water for the valve?	
	Close the valve on the inlet pipe.	
	Open the valve on the pump line.	
	Check the color and quantity of lubricating oil. Is it normal?	
	Open the pump before the inlet opens.	
Operating check	Check the pump vacuum in a full vacuum. Is that normal?	
	Check the electrical condition (voltage and current) under a complete vacuum. Reasonable?	
	Is there a strange noise?	
	Is it normal to check the temperature of the pumps?	
	Check the color and quantity of lubricating oil. Is it normal?	
Check after	After closing the valve on the suction line, continue to pump the pump a few minutes.	
	If you inhale a foreign body, you need to wash it with a cleaning agent.	
	Close the cooling water, and if the pump does not work for a long time, the cooling water in the pump is discharged.	
	Ensure the valve shutdown on the inlet and exhaust pipes.	
	Make sure the power is off.	

6. Vulnerability list

The following are a list of the items used to check the recommended damage items, including all the parts that are sealed, bearing, and other conventions that are not reused. These parts can be ordered by ordering "maintenance packages" and can be purchased individually.

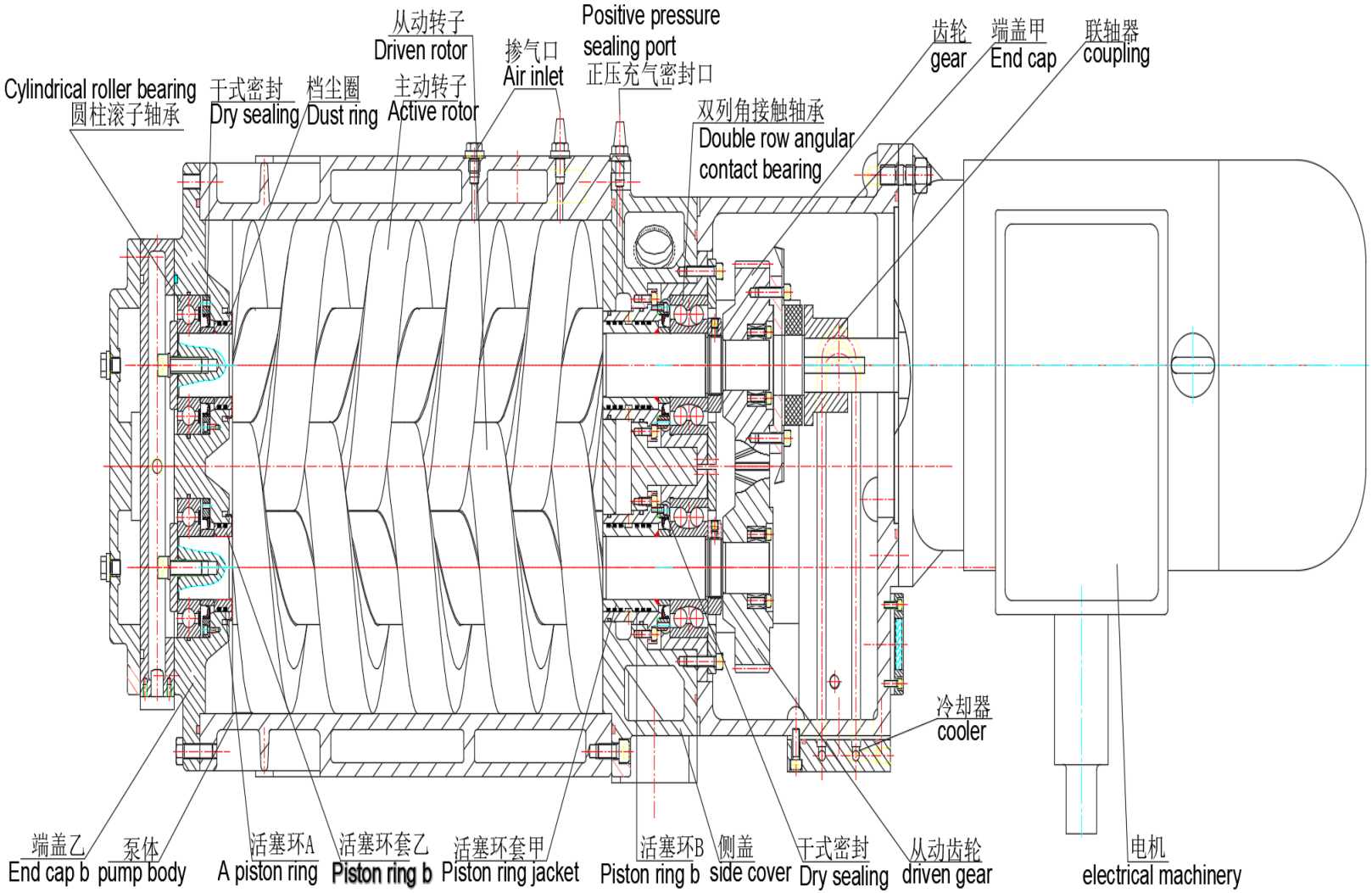
LG\LGB screw vacuum pump vulnerability list:

NO	Class stop	Quantity										
		LG50	LG70	LG100	LG150	LG200	LGB70	LGB110	LGB150	LGB220A	LGB300	LGB300A
1	Piston ring A	14	14	20	20	20	14	14	20	20	20	20
2	Piston ring B	6	6	0	0	0	6	6	0	0	0	0
3	Cylindrical roller bearing	2	2	2	2	2	2	2	2	2	2	2
4	Angle contact ball bearing	2	2	2	2	2	2	2	2	2	2	2
5	Pump O ring	1SET	1SET	1SET	1SET	1SET	1SET	1SET	1SET	1SET	1SET	1SET
6	Side cover O ring	1SET	1SET	1SET	1SET	1SET	1SET	1SET	1SET	1SET	1SET	1SET
7	End head b O ring	1SET	1SET	1SET	1SET	1SET	1SET	1SET	1SET	1SET	1SET	1SET
8	Plug O ring	1SET	1SET	1SET	1SET	1SET	1SET	1SET	1SET	1SET	1SET	1SET
9	Hexagon screw ring	1SET	1SET	1SET	1SET	1SET	1SET	1SET	1SET	1SET	1SET	1SET
10	Motor gasket	1	1	1	1	1	1	1	1	1	1	1
11	Dry seal	4	4	4	4	4	4	4	4	4	4	4
12	Coupling block	1	1	1	1	1	1	1	1	1	1	1

7、外形安装图 External installation drawing

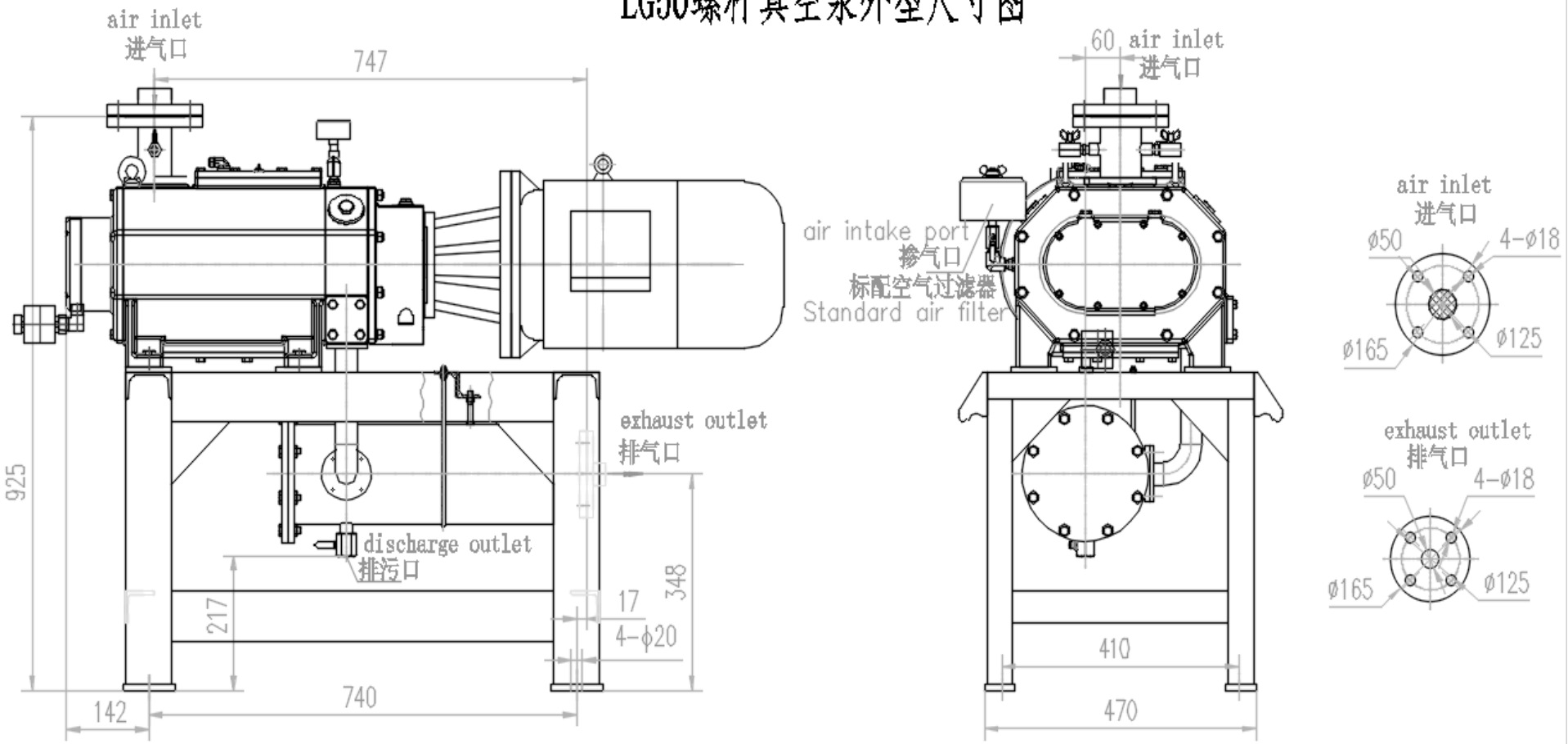
LG 螺杆真空泵结构示意图

LG Structure diagram of screw vacuum pump



LG50 Screw Vacuum Pump Dimensions Diagram

LG50螺杆真空泵外型尺寸图



Single pump performance parameter table

pumping speed	ultimate vacuum	Motor power
50L/S	5Pa	5.5KW
pump speed	inlet diameter	exhaust diameter
2900rpm	50mm	50mm